$$S(w)=\sqrt{(7w)}+e^{-w}$$

$$S(w)=(7w)^{\frac{1}{2}}+e^{-w}$$

$$S'(w)=\frac{1}{2}(7w)^{\frac{1}{2}-\frac{2}{2}}.7+e^{-w}.-1$$

$$S'(w)=\frac{7}{2\sqrt{(7w)}}-e^{-w}$$